

CS 559: Machine Learning: Fundamentals and Applications

Assignment 5 Due: 4/15/2025 Tuesday 11:59 p.m.

- The assignment must be individual work and must not be copied or shared. Any tendency to cheat/copy evidence will lead to a 0 mark for the assignment.
- Students must only use Pandas, NumPy, Matplotlib, and Scipy if the problem does not specify libraries/packages.
- All codes will be tested in grading. Any codes with an error will be marked 0. Make sure to restart the kernel and run it all before the submission. Delete any codes that should not be graded.
- Results must be displayed.
- All problems must be submitted in a single notebook file. Do not use a text editor to write codes.

1 Clustering Analysis [60 pts]

- a. Load the provided data - **autompg.csv**. The data contains seven columns. This data set can be used for both classification and regression problems.
- b. [10 pts] Transform **mpg** into an ordinal categorical feature with three values [good (2), average (1), bad (0)] by calculating the quantiles using **numpy.quantile**. For other continuous columns, convert to four-valued ordinal categorical features as (4) - high, (3) - good, (2) - okay, (1) - poor.
- c. [15 pts] Implement a **myKmeans**(X , k , max_iter) algorithm as a **function** (not a class) **using only NumPy**. The parameter X is the data, k is the number of clusters, and max_iter is the maximum number of iterations. The function returns the final cluster labels, the final centroids of the clusters, and the final **inertia**, which is defined as the sum of the squared distance of each data point to its cluster centroid.
- d. [15 pts] Determine the appropriate k value of the new data set using **myKmeans**. Show that the inertia converges as k increases.
- e. [20 pts] Using the results of **myKmeans** with the best k value from (d), build a linear classification model using Scikit-learn that predicts the cluster label. Split the data into training and test sets by 8 to 2, and report the accuracy for both train and test.

2 Dimensionality Reduction [40 pts]

- a. Load the provided data - **pca.data.csv**.
- b. [15 pts] Study the data and hypothesize the appropriate number of dimensions (n) that the data can be reduced to. Use statistics and visualizations to support your answer.
- c. [15 pts] Implement a function **myPCA**(X , n) (no class) that takes the data X and n is the number of components found in b. The function will return the principal components.
- d. [10 pts] Perform **myPCA** on the data and compare the result with Scikit-learn PCA.